ML1503 – WEB PROGRAMMING (LAB INTEGRATED)

Semester Practical Lab Experiments & Solutions

Direct Question & Concise Solution Manual for All 7 Syllabus Experiments

Experiment 1 HTML5

Question: Design a Webpage using all HTML elements.

exp1/exp1.html

```
<!DOCTYPE html>
<html>
<head>
  <title>All HTML Elements</title>
</head>
<body>
  <h1>Heading 1</h1>
  <h2>Heading 2</h2>

  <p>This paragraph demonstrates <b>bold</b>, <i>italic</i>, and <u>underlined</u> text.</p>

  <a href="https://example.com" target="_blank">Sample Hyperlink</a>
  <br><br>

  <img src="photo.jpg" alt="Sample Image" width="150" height="100">

  <h3>Unordered List</h3>
  <ul>
    <li>HTML</li>
    <li>CSS</li>
    <li>JavaScript</li>
  </ul>

  <h3>Ordered List</h3>
  <ol>
    <li>Node.js</li>
    <li>MongoDB</li>
    <li>AngularJS</li>
  </ol>

  <h3>Table</h3>
  <table border="1" cellpadding="6">
    <tr>
      <th>ID</th>
      <th>Name</th>
      <th>Role</th>
    </tr>
    <tr>
      <td>1</td>
      <td>Alice</td>
      <td>Developer</td>
    </tr>
    <tr>
      <td>2</td>
      <td>Bob</td>
      <td>Designer</td>
    </tr>
  </table>

  <h3>Form Controls</h3>
  <form>
    <label>Text Field: <input type="text" name="username"></label><br><br>

    Gender:
    <label><input type="radio" name="gender" value="m"> Male</label>
    <label><input type="radio" name="gender" value="f"> Female</label><br><br>

    Skills:
    <label><input type="checkbox" name="skill" value="web"> Web</label>
    <label><input type="checkbox" name="skill" value="app"> App</label><br><br>

    <label>City:
      <select name="city">
        <option value="chennai">Chennai</option>
        <option value="bangalore">Bangalore</option>
        <option value="mumbai">Mumbai</option>
      </select>
    </label><br><br>

    <input type="submit" value="Submit Form">
    <button type="button" onclick="alert('Button clicked!')">Click Button</button>
  </form>
</body>
</html>
```

Experiment 2 CSS3

Question: Create a web page with all types of Cascading style sheets and CSS Selectors.

exp2/exp2.html

```
<!DOCTYPE html>
<html>
<head>
  <title>CSS Types and Selectors</title>

  <!-- 1. External CSS -->
  <link rel="stylesheet" href="style.css">

  <!-- 2. Internal CSS -->
  <style>
    /* Element Selector */
    h2 {
      color: darkgreen;
    }
  </style>
</head>
<body>
  <!-- 3. Inline CSS -->
  <h1 style="color: crimson;">1. Inline CSS Heading</h1>

  <!-- Styled via Internal CSS (Element Selector) -->
  <h2>2. Internal CSS Heading (Element Selector)</h2>

  <!-- Styled via External CSS (Class Selector) -->
  <p class="highlight">3. External CSS Paragraph (Class Selector)</p>

  <!-- Styled via External CSS (ID Selector) -->
  <p id="unique">4. External CSS Paragraph (ID Selector)</p>
</body>
</html>
```

exp2/style.css (External Stylesheet)

```
/* Universal Selector */
* {
  font-family: Arial, sans-serif;
}

/* Class Selector */
.highlight {
  background-color: #fff2a8;
  padding: 8px;
  border-left: 4px solid orange;
}

/* ID Selector */
#unique {
  color: royalblue;
  font-weight: bold;
}
```

Experiment 3 JAVASCRIPT & DOM / DHTML

Question: Write Client Side Scripts for Validating Web Form Controls using DHTML.
Design the following using JavaScript and DOM:

a) Include Image Slide Show and Digital clock.

b) Develop a web application to implement online quiz system.

Part 1: Client Side Scripts for Validating Web Form Controls using DHTML

exp3/3_validation.html

```
<!DOCTYPE html>
<html>
<head>
  <title>DHTML Form Validation</title>
  <style>
    .error {
      color: red;
      font-size: 13px;
    }
  </style>
</head>
<body>
  <h2>Web Form Validation (DHTML)</h2>

  <form onsubmit="return validate()">
    <label>Name: <input id="name"></label>
    <span id="nameErr" class="error"></span><br><br>

    <label>Email: <input id="email" type="email"></label>
    <span id="emailErr" class="error"></span><br><br>

    <label>Password: <input id="pass" type="password"></label>
    <span id="passErr" class="error"></span><br><br>

    <button type="submit">Submit</button>
  </form>

  <script>
    function validate() {
      let valid = true;
      let name = document.getElementById("name").value.trim();
      let email = document.getElementById("email").value.trim();
      let pass = document.getElementById("pass").value;

      document.getElementById("nameErr").textContent = name ? "" : "Name is required";
      document.getElementById("emailErr").textContent = /\S+@\S+\.\S+/.test(email) ? "" : "Enter valid email";
      document.getElementById("passErr").textContent = pass.length >= 6 ? "" : "Password must be at least 6 characters";

      if (!name || !/\S+@\S+\.\S+/.test(email) || pass.length < 6) {
        valid = false;
      }

      if (valid) {
        alert("Form validated and submitted successfully!");
      }
      return valid;
    }
  </script>
</body>
</html>
```

Part 3(a): Include Image Slide Show and Digital Clock (using JavaScript and DOM)

exp3/3a_slideshow_clock.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Slideshow and Digital Clock</title>
</head>
<body>
  <h2>Digital Clock</h2>
  <h3 id="clock"></h3>

  <h2>Image Slideshow</h2>
  <img id="slide" src="img1.jpg" width="400" height="250" alt="Slider image">
  <br><br>
  <button onclick="change(-1)">Previous</button>
  <button onclick="change(1)">Next</button>

  <script>
    // Digital Clock
    function updateClock() {
      let d = new Date();
      document.getElementById("clock").textContent = d.toLocaleTimeString();
    }
    updateClock();
    setInterval(updateClock, 1000);

    // Image Slideshow
    let images = ["img1.jpg", "img2.jpg", "img3.jpg"];
    let i = 0;

    function change(step) {
      i = (i + step + images.length) % images.length;
      document.getElementById("slide").src = images[i];
    }

    // Auto slide every 3 seconds
    setInterval(function() {
      change(1);
    }, 3000);
  </script>
</body>
</html>
```

Part 3(b): Online Quiz System (using JavaScript and DOM)

exp3/3b_quiz.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Online Quiz System</title>
</head>
<body>
  <h2>Online Quiz System</h2>

  <form id="quizForm">
    <p><b>1. Which language runs in a web browser?</b></p>
    <label><input type="radio" name="q1" value="Java"> Java</label><br>
    <label><input type="radio" name="q1" value="C"> C</label><br>
    <label><input type="radio" name="q1" value="JavaScript"> JavaScript</label><br>

    <p><b>2. What does CSS stand for?</b></p>
    <label><input type="radio" name="q2" value="Cascading Style Sheets"> Cascading Style Sheets</label><br>
    <label><input type="radio" name="q2" value="Creative Style System"> Creative Style System</label><br>
    <br>

    <button type="button" onclick="submitQuiz()">Submit Quiz</button>
  </form>

  <h3 id="score"></h3>

  <script>
    function submitQuiz() {
      const answers = {
        q1: "JavaScript",
        q2: "Cascading Style Sheets"
      };

      let score = 0;
      let form = document.getElementById("quizForm");

      for (let q in answers) {
        if (form.elements[q].value === answers[q]) {
          score++;
        }
      }

      document.getElementById("score").textContent =
        "Your Score: " + score + " / 2";
    }
  </script>
</body>
</html>
```

Experiment 4 ANGULARJS

Question: Use built-in Angular directives to show and hide elements and display lists of data.

exp4/exp4.html

```
<!DOCTYPE html>
<html ng-app="app">
<head>
  <title>AngularJS Directives</title>
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>
</head>
<body ng-controller="Ctrl">
  <h2>Angular Directives: Show, Hide, and Lists</h2>

  <!-- ng-model -->
  <label>Enter Name: <input type="text" ng-model="name"></label>
  <p>Hello, {{name}}!</p>

  <!-- ng-click and ng-show / ng-hide -->
  <button ng-click="visible = !visible">Toggle Visibility</button>
  <p ng-show="visible" style="color: green;">ng-show: This element is visible.</p>
  <p ng-hide="visible" style="color: red;">ng-hide: This element is hidden.</p>

  <!-- ng-repeat -->
  <h3>List of Subjects (ng-repeat)</h3>
  <ul>
    <li ng-repeat="x in items">{{x}}</li>
  </ul>

  <script>
    var app = angular.module("app", []);
    app.controller("Ctrl", function($scope) {
      $scope.name = "Student";
      $scope.visible = true;
      $scope.items = ["HTML5", "CSS3", "JavaScript", "AngularJS", "Node.js"];
    });
  </script>
</body>
</html>
```

Experiment 5 ANGULARJS

Question: Design a shopping cart application using AngularJS. Your shopping webpage should have the provisions for selecting the list of items from different category, Once the items are selected on clicking the submit button the items in the cart with its price should be displayed.

exp5/exp5.html

```
<!DOCTYPE html>
<html ng-app="cartApp">
<head>
  <title>AngularJS Shopping Cart</title>
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>
  <style>
    body { font-family: Arial, sans-serif; }
    table { border-collapse: collapse; width: 360px; margin-top: 10px; }
    th, td { border: 1px solid #aaa; padding: 6px; text-align: left; }
  </style>
</head>
<body ng-controller="CartCtrl">
  <h2>Shopping Cart Application</h2>

  <h3>Select Items by Category</h3>
  <div ng-repeat="p in products">
    <label>
      <input type="checkbox" ng-model="p.selected">
      <b>[{{p.category}}]</b> {{p.name}} - Rs. {{p.price}}
    </label>
  </div>
  <br>

  <button ng-click="submitCart()">Submit</button>

  <div ng-show="submitted">
    <h3>Cart Items</h3>
    <table ng-if="cart.length > 0">
      <tr>
        <th>Category</th>
        <th>Item</th>
        <th>Price (Rs.)</th>
      </tr>
      <tr ng-repeat="item in cart">
        <td>{{item.category}}</td>
        <td>{{item.name}}</td>
        <td>{{item.price}}</td>
      </tr>
      <tr>
        <th colspan="2">Total Price</th>
        <th>Rs. {{total}}</th>
      </tr>
    </table>
    <p ng-if="cart.length === 0">No items selected.</p>
  </div>

  <script>
    angular.module("cartApp", []).controller("CartCtrl", function($scope) {
      $scope.products = [
        { category: "Food", name: "Rice", price: 60, selected: false },
        { category: "Food", name: "Milk", price: 30, selected: false },
        { category: "Stationery", name: "Pen", price: 10, selected: false },
        { category: "Stationery", name: "Notebook", price: 40, selected: false },
        { category: "Electronics", name: "Mouse", price: 350, selected: false }
      ];

      $scope.cart = [];
      $scope.total = 0;
      $scope.submitted = false;

      $scope.submitCart = function() {
        $scope.cart = $scope.products.filter(p => p.selected);
        $scope.total = $scope.cart.reduce((sum, p) => sum + p.price, 0);
        $scope.submitted = true;
      };
    });
  </script>
</body>
</html>
```

Experiment 6 NODE.JS, MONGODB & ANGULARJS

Question: Design a Command Line Application for an online super market using Node JS and MongoDB.
a) Perform a search based on product id or name.
b) On retrieving the results, display the product details of different brands in table format with the Price field in sorted order using AngularJS.

Part 6(a): Command Line Application using Node.js and MongoDB (Search by ID or Name)

`exp6/6a_search.js`

```
// Prerequisites:
// npm install mongodb
//
// MongoDB Setup Commands in mongosh:
// use supermarket;
// db.products.insertMany([
//   { id: 1, name: "Soap", brand: "Dove", price: 45 },
//   { id: 2, name: "Soap", brand: "Lux", price: 30 },
//   { id: 3, name: "Rice", brand: "IndiaGate", price: 90 },
//   { id: 4, name: "Rice", brand: "Daawat", price: 80 }
// ]);

const { MongoClient } = require("mongodb");
const readline = require("readline");

const url = "mongodb://localhost:27017";
const client = new MongoClient(url);

const rl = readline.createInterface({
  input: process.stdin,
  output: process.stdout
});

async function main() {
  try {
    await client.connect();
    const collection = client.db("supermarket").collection("products");

    rl.question("Enter product ID or name to search: ", async (query) => {
      const isNum = !isNaN(query) && query.trim() !== "";
      const filter = isNum
        ? { id: Number(query) }
        : { name: { $regex: query, $options: "i" } };

      const results = await collection.find(filter).toArray();
      console.log("\n--- Search Results ---");
      console.table(results);

      rl.close();
      await client.close();
    });
  } catch (err) {
    console.error("Error:", err.message);
    rl.close();
  }
}

main();
```

Part 6(b): Display Product Details in Table Sorted by Price using AngularJS

exp6/6b_table.html

```
<!DOCTYPE html>
<html ng-app="marketApp">
<head>
  <title>Supermarket Products Sorted</title>
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>
  <style>
    body { font-family: Arial, sans-serif; }
    table { border-collapse: collapse; width: 420px; margin-top: 15px; }
    th, td { border: 1px solid #333; padding: 8px; text-align: left; }
    th { background: #eee; }
  </style>
</head>
<body ng-controller="MarketCtrl">
  <h2>Supermarket Product Details</h2>

  <label>Search: <input type="text" ng-model="query" placeholder="Filter by name/brand"></label>
  <br><br>

  <label>Sort Order:
    <select ng-model="reverse">
      <option ng-value="false">Price: Low to High</option>
      <option ng-value="true">Price: High to Low</option>
    </select>
  </label>

  <!-- Table with Price field in sorted order using AngularJS orderBy -->
  <table>
    <tr>
      <th>Product ID</th>
      <th>Product Name</th>
      <th>Brand</th>
      <th>Price (Rs.)</th>
    </tr>
    <tr ng-repeat="p in products | filter:query | orderBy:'price':reverse">
      <td>{{p.id}}</td>
      <td>{{p.name}}</td>
      <td>{{p.brand}}</td>
      <td>{{p.price}}</td>
    </tr>
  </table>

  <script>
    angular.module("marketApp", []).controller("MarketCtrl", function($scope) {
      $scope.reverse = false;

      // Product details of different brands retrieved from database
      $scope.products = [
        { id: 101, name: "Soap", brand: "Dove", price: 55 },
        { id: 102, name: "Soap", brand: "Lux", price: 32 },
        { id: 103, name: "Soap", brand: "Pears", price: 48 },
        { id: 104, name: "Rice", brand: "Daawat", price: 75 },
        { id: 105, name: "Rice", brand: "India Gate", price: 95 },
        { id: 106, name: "Shampoo", brand: "Pantene", price: 120 },
        { id: 107, name: "Shampoo", brand: "Sunsilk", price: 85 }
      ];
    });
  </script>
</body>
</html>
```

Experiment 7 EXPRESS.JS & AJAX

Question: To Build an
a) API by following REST syntax and Express JS and AJAX Application.

Part 7(a) Server: REST API using Express.js

exp7/server.js

```
// Prerequisites:
// npm install express cors
// Run server: node server.js

const express = require("express");
const cors = require("cors");
const app = express();

app.use(cors());
app.use(express.json());

// In-memory data store
let products = [
  { id: 1, name: "Laptop", price: 55000 },
  { id: 2, name: "Mouse", price: 450 },
  { id: 3, name: "Keyboard", price: 1100 }
];

// REST API: GET all products
app.get("/api/products", (req, res) => {
  res.json(products);
});

// REST API: POST create a new product
app.post("/api/products", (req, res) => {
  const item = {
    id: products.length + 1,
    name: req.body.name,
    price: Number(req.body.price)
  };
  products.push(item);
  res.status(201).json(item);
});

app.listen(3000, () => {
  console.log("REST API Server running at http://localhost:3000");
});
```

Part 7(a) Client: AJAX Application consuming the REST API

exp7/client.html

```
<!DOCTYPE html>
<html>
<head>
  <title>REST API + AJAX Application</title>
  <style>
    body { font-family: Arial, sans-serif; }
    table { border-collapse: collapse; width: 360px; margin-top: 15px; }
    th, td { border: 1px solid #ccc; padding: 6px; text-align: left; }
    th { background: #f4f4f4; }
  </style>
</head>
<body>
  <h2>REST API and AJAX Application</h2>

  <h3>Add Product (POST via AJAX)</h3>
  <input id="name" placeholder="Product Name">
  <input id="price" type="number" placeholder="Price">
  <button onclick="addProduct()">Add</button>

  <h3>Product List (GET via AJAX)</h3>
  <button onclick="getProducts()">Refresh Products</button>

  <table>
    <thead>
      <tr>
        <th>ID</th>
        <th>Product Name</th>
        <th>Price (Rs.)</th>
      </tr>
    </thead>
    <tbody id="tableBody"></tbody>
  </table>

  <script>
    const API = "http://localhost:3000/api/products";

    // AJAX GET request
    async function getProducts() {
      try {
        const res = await fetch(API);
        const data = await res.json();
        const tbody = document.getElementById("tableBody");
        tbody.innerHTML = "";

        data.forEach(p => {
          tbody.innerHTML += `<tr><td>${p.id}</td><td>${p.name}</td><td>${p.price}</td></tr>`;
        });
      } catch (err) {
        alert("Make sure server.js is running on http://localhost:3000");
      }
    }

    // AJAX POST request
    async function addProduct() {
      const name = document.getElementById("name").value.trim();
      const price = document.getElementById("price").value.trim();
      if (!name || !price) {
        alert("Please enter both product name and price");
        return;
      }

      await fetch(API, {
        method: "POST",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify({ name, price })
      });

      document.getElementById("name").value = "";
      document.getElementById("price").value = "";
      getProducts();
    }

    // Initial load
    getProducts();
  </script>
</body>
</html>
```